Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/24/2006
PI Name:	Hines, John M.S.		
Project Title:	Development of NASA-Specific Bioinformatics Environment		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced environmental monitoring and control		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Project Type:	Ground	**Solicitation / Funding Source:**	INTERNAL PROJECT
Start Date:	09/30/2001	**End Date:**	09/30/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Schweighofer, Karl (USRA/RIACS) Shrager, Jeff (ISLE)		
Grant/Contract No.:			
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Performance Goal Text:			
	Bioinformatics is the application of computational and theoretical tools to the analysis and interpretation of biological data. Bioinformatics is essential for management and analysis of data in modern biology and medicine. The main objective of the proposed work is to develop bioinformatics technology capabilities that will help meet unique requirements related to biological experiments both on future flight missions and on earth. The work has two specific goals: 1) To design, build, deploy, and maintain a bioinformatics facility needed for solving NASA-specific biology problems. This facility will consist of tools, databases and knowledge management systems suited for NASA goals. It will provide a unified and integrated platform for NASA scientists and their collaborators to share data, information and analyses. Such a facility will greatly improve the capabilities for scientific discovery, and will contribute toward the understanding of how genomic and environmental factors drive evolution and impact on astronaut health during long-term space travel. The full suite of bioinformatics resources will be available freely to the NASA community.		

Task Description:	2) To enhance the capabilities of the facility by developing novel computational technologies for analyzing and integrating new biological data and already existing knowledge, and by integrating bioinformatics capabilities with massively parallel computational environment at NASA-Ames. NASA's interests in bioinformatics and computational biology center around two broad and well-established research areas: 1. Understanding the parameters for life and how organisms adapt to extreme conditions. 2. Gaining a better understanding of the effects of long-duration space travel on humans. The first of these is important to understand how essential cellular processes arose, evolved and adapted to diverse conditions on earth. It helps us to understand how interactions between living organisms and their environment shaped the early earth and gave rise to biodiversity on our planet. It will also help us to formulate the criteria for life to exist on other planets. The second research program is directed at a better understanding of the effects of long-duration space travel on humans. There are many known adverse effects due to prolonged stays in space. These include the well known effects of bone loss, muscle degeneration, and immune suppression. Such conditions are large-scale observables resulting from altered physiology in response to a unique stress environment. However, there may be many other effects, present at the molecular level, which go undetected because they do not generate an obvious short-term phenotypic effect. These factors could seriously impact our ability to take part in extended manned missions (for example, to Mars). It is therefore essential that we detect these changes at the molecular level and develop effective countermeasures. In addition, since some effects of the space environment mimic the effects of aging and disease, studies on these effects could impact terrestrial medicine by leading to new treatments for osteoporosis and other diseases associated with the aging process. To begin to address these issues, we turn to the tools of modern molecular biology, and we dedicate ourselves to acquiring the data necessary for understanding how genes are regulated and respond to changes in the environment. However, interpreting these data and developing the causal links between gene expression and phenotypic changes in organisms is not possible without sophisticated tools of bioinformatics. The proposed bioinformatics facility will provide these tools to the whole community in a form suited for solving NASA specific problems. By doing so, it will considerably increase potential for scientific discoveries in the field. Although several facilities in national laboratories and universities, such as NCBI, TIGR, San Diego Supercomputer Workbench or Stanford Microarray Database provide bioinformatics capabilities that might be suitable for occasional individual user, they are insufficient for the NASA community. Our will have crucial capabilities not available elsewhere. Examples are: 1) All bioinformatics tools needed to support biological research will be present in a single site and interfaced with each other, thus eliminating the need for difficult transformations of data. 2) The facility will contain databases and knowledge bases developed for NASA. For example, genomic, proteomic, metabolic and transcription factors databases for microorganism that are of interest will be compiled, maintained and constantly updated This will greatly improve both individual and collaborative research. 3) The facility will provide capabilities to handle problems that are computationally or memory intensive. Reconstructions of phylogenetic relationships and refined analyses of microarray data fall into this category. 4) Novel tools developed by NASA scientists and their collaborators will be available. Some of these tools, aimed at reconstructing regulatory networks in cells, are described further in this proposal. 5) The bioinformatics staff will work with experimental biologists to provide them with computational capablities needed so solve their specific problems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	One area of research is directed at a better understanding of the effects of long-duration space travel on humans. There are many known adverse effects due to prolonged stays in space. These include the well known effects of bone loss, muscle degeneration, and immune suppression. Such conditions are large-scale observables resulting from altered physiology in response to a unique stress environment. However, there may be many other effects, present at the molecular level, which go undetected because they do not generate an obvious short-term phenotypic effect. These factors could seriously impact our ability to take part in extended manned missions (for example, to Mars). It is therefore essential that we detect these changes at the molecular level and develop effective countermeasures. In addition, since some effects of the space environment mimic the effects of aging and disease, studies on these effects could impact terrestrial medicine by leading to new treatments for osteoporosis and other diseases associated with the aging process.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)